

LOOKS THE SAME, ANSWERS DIFFERENTLY: FLIP-DIRECTION STEERING FOR ROBUST VISION- LANGUAGE REASONING

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ABSTRACT

Vision-language models (VLMs) achieve strong visual reasoning performance, yet subtle changes from routine image capture and processing can alter their reasoning trajectories even when images appear nearly identical. In long-horizon generation, the resulting activation shifts may accumulate across decoding steps, progressively altering reasoning tokens and ultimately changing the final answer, a phenomenon referred to as *answer flips*. To address this instability, we propose FLIPDIR (Flip-Direction Steering), a training-free inference-time method that estimates a low-rank flip-inducing activation subspace from contrastive pairs of original and answer-flipping inputs and selectively steers hidden states during decoding. A margin-based gate limits subspace attenuation to uncertain decoding steps, recovering original predictions while preserving stable ones. To evaluate this balance, we introduce VISFLIP, pairing original, flip, and non-flip inputs across scientific reasoning, robot-scene understanding, and medical VQA. Its metrics separate consistency with each model’s original predictions from underlying task accuracy. Across 18 settings, FLIPDIR improves the harmonic mean of clean preservation, non-flip preservation, and flip recovery by 2.8 points on average and up to 6.3 points over the strongest baseline per setting. We will publicly release the complete benchmark, its construction code, and all experimental code.

1 INTRODUCTION

Recent vision-language models (VLMs) achieve strong performance across scientific reasoning, robot-scene understanding, and medical visual question answering (Zhang et al., 2024; Chen et al., 2025; Li et al., 2023a). Despite these advances, generating reliable and consistent responses to diverse visual inputs in real-world scenarios remains challenging (Rosenfeld et al., 2025). This requires models to preserve task-relevant visual information across image variations and integrate it consistently with the textual context throughout generation. These challenges have motivated research on several aspects of VLM reliability, including adversarial robustness (Xie et al., 2025), multimodal safety (Wang et al., 2025b), and visual hallucination (Yang et al., 2025). Recent studies further show that, even without deliberate attacks, benign corruptions such as additive noise can degrade VLM performance (Qiu et al., 2025).

However, common real-world scenarios involve benign inputs with subtle visual variations rather than deliberate attacks, unsafe requests, or pronounced image degradation. Although safety and resistance to adversarial prompts receive considerable attention in recent post-training (Gemma Team et al., 2025; Meta, 2025), the effects of subtle visual variations, such as slight exposure shifts or JPEG compression, on long-horizon visual reasoning remain underexplored. As shown in Table 1, we observe that subtle exposure shifts can change final predictions even in frontier models, a phenomenon we call *answer flips*. On M3CoT (Chen et al., 2024b), a benchmark for multi-step multimodal reasoning, flip rates are 9.6% for GPT-5.5 and 21.7% for Claude Opus 4.8 (OpenAI, 2026;

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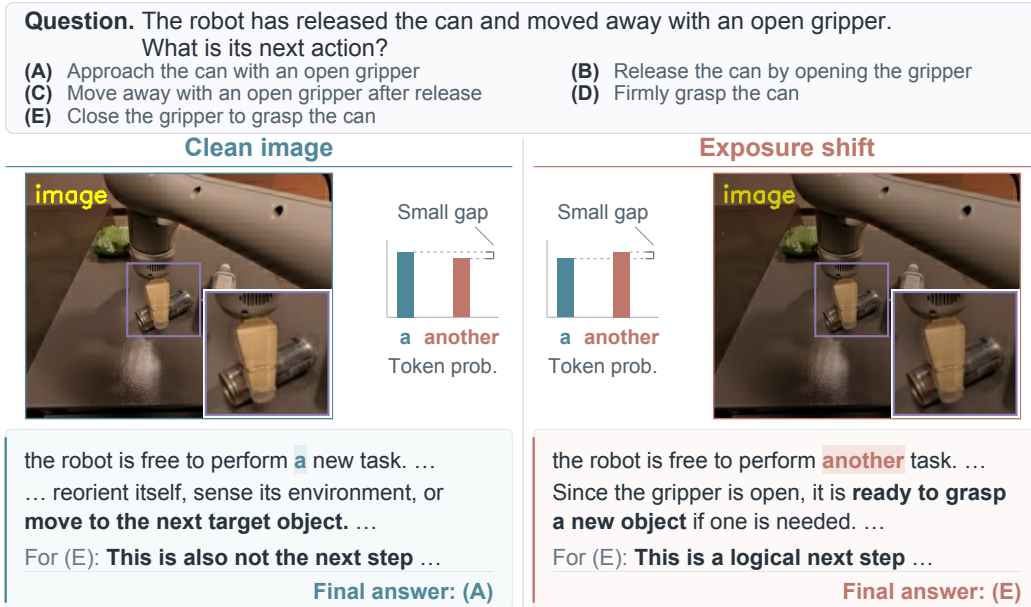


Figure 1: A **subtle exposure shift can flip the final answer**. For nearly identical Robo2VLM images, Qwen3-VL-8B produces identical text before diverging at *a* versus *another* and reaching different action predictions. Token probabilities are schematic.

Anthropic, 2026) (details in Sec. 4 and Appendix B.2). Figure 1 shows an *answer flip* under a subtle exposure shift, where initially identical generations first differ at *a new task* versus *another task* and later reach different action predictions. Moreover, these early token changes can remain locally plausible, making it difficult to identify when intervention is needed.

Table 1: **Answer flips on recent models.**

	GPT-5.5	Claude Opus 4.8	Qwen3-VL-8B	Gemma-3-12B
Clean accuracy (%)	93.2	91.8	68.3	47.8
Answer flip (%)	9.6	21.7	47.6	62.4

To address this challenge, we propose FLIPDIR (Flip-Direction Steering), a training-free inference-time method that estimates a low-rank flip-inducing activation subspace and uses it to selectively steer hidden states during decoding. We estimate this subspace offline by applying singular value decomposition (SVD) to activation differences from a set of contrastive pairs of original and answer-flipping inputs. During inference on new inputs, a margin-based gate uses the gap between the two largest next-token probabilities to select uncertain decoding steps for intervention (Figure 5). When the gate opens, we attenuate the part of the current hidden state that lies in the estimated subspace; otherwise, we retain the unedited next-token distribution. Calibration requires no gradient updates or auxiliary model training, and inference uses the stored basis without a corresponding original image.

To enable systematic evaluation of recovery and preservation under subtle visual variations, we introduce VISFLIP, covering scientific reasoning, robot-scene understanding, and medical visual question answering. Across nine dataset-perturbation combinations, we sample low-intensity perturbations that reflect visual changes commonly encountered in each task’s real-world setting. The benchmark pairs original inputs with flip and non-flip variants, with fixed sampling ranges and recorded seeds and parameters for reproducible image generation (details in Section 4 and Appendix A.2). Its evaluation measures clean preservation, non-flip preservation, and flip recovery relative to each model’s original predictions, separating perturbation-induced instability from each model’s underlying task accuracy.

Across 18 settings, FLIPDIR improves the harmonic mean of these metrics by 2.8 points on average and up to 6.3 points over the strongest baseline in each setting. Ablation studies show that subspace estimation from contrastive pairs and margin-based gating both improve the balance between recov-

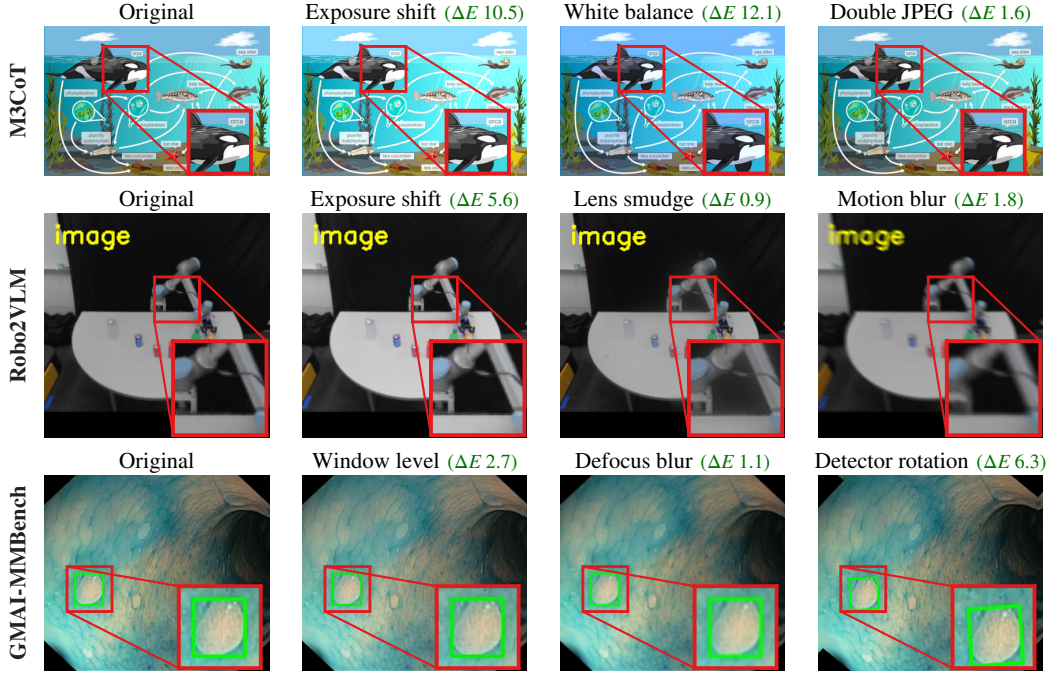


Figure 2: **Visual variations in VISFLIP.** Rows show subtle variations common to each domain. ΔE denotes the mean per-pixel color difference from the original image (parameter ranges in Appendix A.2).

ery and preservation. We will publicly release the complete benchmark, a reproducible construction and evaluation framework for additional models and perturbation settings, and all experimental code.

Our contributions are threefold.

- We analyze *answer flips* under subtle real-world visual variations, examining the associated activation shifts and decoding divergence.
- We propose FLIPDIR, a training-free inference-time method that estimates a low-rank flip-inducing activation subspace and uses margin-based gating to selectively steer hidden states during decoding.
- We introduce VISFLIP, a benchmark for answer recovery and preservation under subtle variations from routine image capture and processing, spanning scientific reasoning, robot-scene understanding, and medical VQA.

2 RELATED WORK

Robustness to real-world visual perturbations. Prior work studies natural corruptions, acquisition shifts, and semantics-preserving transformations (Hendrycks & Dietterich, 2019; Di Salvo et al., 2024). VLM evaluations also examine adversarial perturbations, hallucination, and aggregate performance under input variations (Zhao et al., 2023; Qi et al., 2024; Li et al., 2023b). We focus on the clean-to-flip transition under mild, naturally occurring perturbations and on recovering the model’s clean prediction.

Jailbreaking and safety steering. Refusal can be mediated by a single activation direction (Arditi et al., 2024), while multimodal defenses use activation differences associated with adversarial visual features to suppress harmful responses (Wang et al., 2025a). Such methods target an identifiable behavior. In our setting, the same perturbation may change or preserve the answer, so the useful contrast must be identified from its effect on the model’s prediction.

Hallucination mitigation. Decoding methods improve visual grounding through visual contrast, token dependencies, or uncertainty (Leng et al., 2024; Huang et al., 2024; Xu et al., 2026);

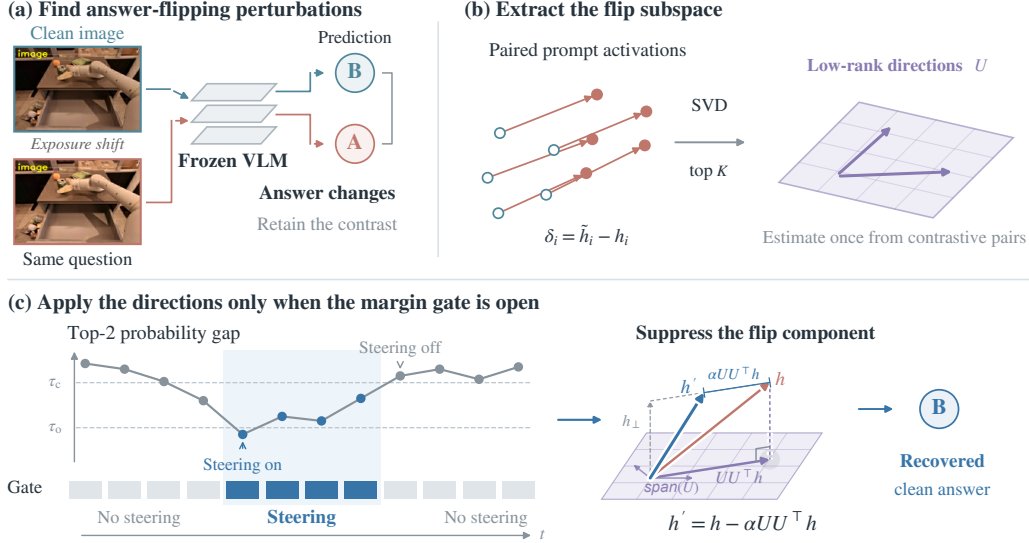


Figure 3: **FLIPDIR overview.** (a) Identify *answer flips*. (b) Estimate U via SVD of paired activation differences. (c) Attenuate the subspace projection at margin-gated decoding steps. The plane denotes $\text{span}(U)$, and increasing α moves h' toward $h_{\perp} = (I - UU^T)h$.

activation-based methods steer representations toward grounded behavior (Liu et al., 2025). Nullu identifies hallucination-associated subspaces by contrasting truthful and hallucinated descriptions of the same visual content (Yang et al., 2025). FLIPDIR instead derives its subspace from prediction-changing visual perturbations and applies it selectively at uncertain decoding steps.

3 FLIPDIR: SELECTIVE STEERING FOR ANSWER-FLIP RECOVERY

FLIPDIR estimates a low-rank flip-inducing activation subspace from activation differences between original inputs and their answer-flipping variants. During inference on new inputs, a margin-based gate selects decoding steps at which to attenuate the part of the hidden state that lies in this subspace (Figure 3). Subspace estimation is performed offline using a set of contrastive pairs; inference reuses the resulting basis without gradient updates or auxiliary model training.

3.1 PROBLEM SETUP

We study answer instability under mild visual changes that leave images nearly identical, such as exposure shifts, blur, and JPEG recompression. The objective is to recover the model’s original answer when a perturbation changes it, while preserving answers on unaffected inputs. Let f be a VLM with original answer $y = f(I, T)$ for image I and prompt T . Let $\mathcal{P}$ denote the sampling distribution over transformations in a predefined perturbation family. For $\tau \sim \mathcal{P}$, let $\tilde{I} = \tau(I)$. A perturbation is *flip-inducing* when

$$\text{norm}(f(\tilde{I}, T)) \neq \text{norm}(f(I, T)), \quad (1)$$

where $\text{norm}(\cdot)$ maps the model output to the task-specific answer space. Flip labels are model-specific because they reference the model’s clean prediction. Calibration is performed separately for each model, dataset, and known perturbation family; the resulting intervention is reused on new inputs from that setting without a clean reference or advance knowledge of their flip status.

3.2 ESTIMATING THE FLIP-INDUCING SUBSPACE

For each model and perturbation setting, let $\mathcal{D}_{\text{flip}} = \{(I_i, \tilde{I}_i, T_i)\}_{i=1}^n$ be a set of contrastive pairs satisfying Eq. 1, with at most one pair per original input and $n = |\mathcal{D}_{\text{flip}}|$. This selection isolates answer-changing perturbations within families that can also preserve predictions.

Paired activation differences. Let $h_{\ell}(I, T) \in \mathbb{R}^d$ be the hidden state at layer ℓ for the final prompt token after multimodal prefill. We extract this state from unedited passes on the clean and flipped

inputs and compute

$$\delta_i = h_\ell(\tilde{I}_i, T_i) - h_\ell(I_i, T_i), \quad \tilde{I}_i = \tau_i(I_i). \quad (2)$$

Thus, δ_i measures the paired activation shift at the final prompt-token position. We stack the differences from n pairs into $\Delta = [\delta_1, \dots, \delta_n]^\top \in \mathbb{R}^{n \times d}$.

Subspace estimation. We approximate these activation differences with a k -dimensional subspace. An orthonormal basis U that minimizes the reconstruction error is obtained by

$$\min_{U^\top U = I_k} \|\Delta - \Delta U U^\top\|_F^2, \quad U = V_{:,1:k} \in \mathbb{R}^{d \times k}, \quad (3)$$

where V contains the right singular vectors of $\Delta = P\Sigma V^\top$, ordered by decreasing singular value. Thus, U retains k singular directions, with $1 \leq k \leq \text{rank}(\Delta) \leq \min(n, d)$. We use uncentered SVD so that the mean difference, which may contain a shared clean-to-flip shift, contributes to the estimated subspace. The basis is computed once from these contrastive pairs and fixed during inference. We call $\mathcal{S}_{\text{flip}} = \text{span}(U) \subseteq \mathbb{R}^d$ the estimated flip-inducing subspace. It summarizes activation shifts under answer-flipping perturbations and may also include other perturbation-related variation.

Spectral concentration. We measure the fraction of squared activation differences captured by the top k directions as $E(k) = \sum_{j=1}^k \sigma_j^2 / \sum_j \sigma_j^2$, where σ_j are the singular values. For 255 Qwen3-VL-8B/Robo2VLM lens-smudge pairs ($d = 4096$), Figure 4 compares layers 12 and 36 with a Gaussian null; layer 4 is shown in Appendix D.1 (Figure 10a). The leading direction alone captures 15.6% to 21.2% of the energy, compared with 0.6% for the null, and energy accumulates rapidly at small ranks. In this setting, the spectrum suggests that a compact subspace can capture a substantial fraction of the observed activation changes. Paired differences may also contain image- and prompt-specific variation, so retaining more directions need not improve steering on new inputs. We evaluate the effect of rank on steering performance in Section 5.4.

3.3 MARGIN-GATED STEERING

At inference, we use the estimated basis U to attenuate the hidden-state component within the subspace. We apply this update selectively, using the next-token margin to determine when to intervene.

Hidden-state steering. Let $h \in \mathbb{R}^d$ be the hidden state of the current token at the selected layer ℓ . We steer this state by attenuating its projection onto the estimated subspace:

$$h' = h - \alpha U U^\top h = (I - \alpha U U^\top)h + (1 - \alpha) U U^\top h. \quad (4)$$

The update scales the projection $U U^\top h$ by $1 - \alpha$, leaving its orthogonal component unchanged. Thus, $\alpha = 1$ removes the projection, while $0 < \alpha < 1$ partially attenuates it. We then recompute the remaining layers and output head to obtain the next-token distribution. We edit only decoding states; prompt prefill is unchanged. During inference, steering uses only the current hidden state and stored basis, without a corresponding clean image, activation, or answer. Section 5.1 and Appendix C.2 detail the experimental settings and validation-based selection of ℓ and α .

Next-token margin. Applying this update at every decoding step can also alter stable predictions. We therefore use the model’s next-token uncertainty to select when to intervene. At decoding step t , we first run an unedited forward pass conditioned on the tokens generated so far. Let $p_t^{(1)}$ and $p_t^{(2)}$ be the two largest next-token probabilities. Their difference, $m_t = p_t^{(1)} - p_t^{(2)}$, is small when the model assigns similar probabilities to competing tokens. To test its association with *answer flips*, we analyze stored unedited generation traces. For flipped cases, we label differing next-token predictions under a shared prefix as *divergence tokens*, realigning the prefix after each mismatch; matching predictions are *non-divergence tokens*. Across 7 Gemma and 8 Qwen settings (2,775 divergence and 88,071 non-divergence tokens), divergence tokens have lower median margins in every setting. Pooled over tokens, 93.0% of divergence tokens have $m_t < 0.3$, compared with 11.7% of non-divergence tokens (Figure 5 for Qwen; Appendix D.2, Figure 10b for Gemma). This association motivates using a low margin to trigger steering. The original input is used only to label tokens for this analysis; the gate does not require it at inference.

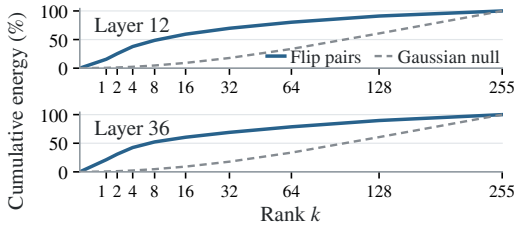


Figure 4: **Low-rank concentration.** Leading directions capture more energy in comparison with a Gaussian null.

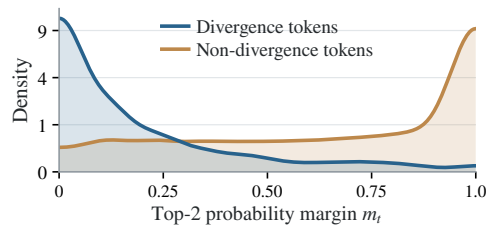


Figure 5: **Next-token margins.** Divergence tokens tend to have lower margins in Qwen3-VL-8B.

Margin gate. With $\tau_o < \tau_c$, the stateful gate opens when $m_t < \tau_o$, closes when $m_t > \tau_c$, and retains its previous state otherwise. Gate decisions always use the unedited probabilities, so they do not depend on confidence introduced by the current edit. When the gate is open, we apply Eq. 4; otherwise, we use the original next-token distribution.

4 VISFLIP

VISFLIP studies *answer flips* under three mild real-world perturbations per domain across science, robotics, and medical VQA (Figure 2).

Science reasoning. We use the physics subset of M3CoT (Chen et al., 2024b), which requires multi-step reasoning over scientific diagrams and scenes. We consider exposure shift, white-balance shift, and double JPEG compression, covering changes in illumination, camera color processing, and image recompression during capture and transmission (Hendrycks & Dietterich, 2019; Afifi et al., 2019).

Robot scenes. We use Robo2VLM (Chen et al., 2025), which covers spatial relations, robot state, and action outcomes in manipulation scenes. We consider exposure shift, lens contamination, and motion blur, reflecting lighting changes, contamination of robot cameras, and camera or platform motion (Hendrycks & Dietterich, 2019; Park et al., 2026).

Medical VQA. We use GMAI-MMBench (Chen et al., 2024a), which spans clinical VQA tasks, perceptual granularities, and imaging modalities. We consider window-level variation, modality-specific defocus blur, and small detector rotation, representing variation in image display, optical or detector focus, and acquisition geometry (Bushberg et al., 2020; Goodman, 2005; Mannan & Langer, 2016).

Table 2 lists parent-image counts by split and representative tasks or modalities. GMAI-MMBench splits are stratified by clinical task and perceptual granularity, preserving these distributions and modality proportions. Pair search requires 12.8 ± 6.4 VLM calls on average, including unsuccessful searches (Table 8). Appendices A.1 through A.4 detail construction and filtering, and Appendix A.5 covers medical preprocessing. We will release the data, fixed perturbation parameters, and image-regeneration code.

Table 2: Composition of VISFLIP across three domains.

Domain	Source	Train	Val	Test	Coverage
Science reasoning	M3CoT	1,582	221	450	Forces & motion, energy, heat, magnetism
Robot scenes	Robo2VLM	1,500	200	567	Robot state, reachability, depth, next action
Medical VQA	GMAI-MMBench	300	200	200	X-ray, CT, endoscopy, fundus

5 EXPERIMENTS

We evaluate recovery, preservation, transfer, and computational cost on VISFLIP using two open-weight VLMs and inference-time baselines.

Table 3: Main results on **Qwen3-VL-8B** (%): clean preservation (C), non-flip preservation (N), flip recovery (F), and their harmonic mean H. Best H is bold; green differences indicate **FLIPDIR**’s gain in H over the strongest baseline.

<i>M3CoT</i>	Exposure				White balance				JPEG			
	C	N	F	H	C	N	F	H	C	N	F	H
Base	100	100	0.0	0.0	100	100	0.0	0.0	100	100	0.0	0.0
VTI	54.1	50.5	33.9	44.2	54.1	56.3	37.9	47.9	54.1	57.7	38.5	48.6
VCD	81.6	84.6	60.7	74.0	83.0	82.7	62.6	74.8	82.3	82.5	55.6	71.0
LEAD	87.6	89.0	62.8	77.8	87.6	87.6	56.3	73.9	87.6	87.4	56.1	73.8
FLIPDIR	87.4	87.9	64.5	78.3 (+0.5)	84.6	84.1	64.2	76.4 (+1.6)	87.2	87.1	66.3	78.9 (+5.1)
<i>Robo2VLM</i>	Exposure				Lens smudge				Motion blur			
	C	N	F	H	C	N	F	H	C	N	F	H
Base	100	100	0.0	0.0	100	100	0.0	0.0	100	100	0.0	0.0
VTI	90.2	90.0	61.7	78.1	90.3	89.9	54.7	74.1	90.2	90.7	55.3	74.6
VCD	91.1	90.5	62.6	79.0	91.9	93.0	66.3	81.7	91.1	89.4	63.8	79.3
LEAD	96.1	94.3	60.7	80.1	96.4	97.1	58.1	79.2	96.1	94.1	47.9	71.6
FLIPDIR	92.5	93.2	67.3	82.4 (+2.3)	94.1	94.0	66.3	82.5 (+0.8)	93.2	93.0	67.0	82.4 (+3.1)
<i>GMAI-MMBench</i>	Window level				Defocus blur				Rotation			
	C	N	F	H	C	N	F	H	C	N	F	H
Base	100	100	0.0	0.0	100	100	0.0	0.0	100	100	0.0	0.0
VTI	45.8	47.4	39.3	43.9	45.8	43.4	32.9	39.9	45.8	42.7	37.3	41.6
VCD	71.4	78.9	59.8	69.1	76.6	72.5	45.9	61.7	72.4	76.0	50.0	63.9
LEAD	85.4	81.1	45.5	65.2	85.4	78.8	40.0	60.7	85.4	84.4	44.5	65.2
FLIPDIR	79.7	84.2	54.5	70.1 (+1.0)	80.2	81.5	44.7	63.7 (+2.0)	80.2	81.8	56.4	70.7 (+5.5)

Table 4: Main results on **Gemma-3-12B**; notation as in Table 3.

<i>M3CoT</i>	Exposure				White balance				JPEG			
	C	N	F	H	C	N	F	H	C	N	F	H
Base	100	100	0.0	0.0	100	100	0.0	0.0	100	100	0.0	0.0
VTI	39.2	34.9	31.7	35.0	39.2	38.8	29.0	35.0	39.2	35.4	33.9	36.0
VCD	66.4	66.2	42.4	55.8	66.1	66.3	48.2	58.9	66.1	67.3	44.3	57.1
LEAD	63.0	66.4	47.8	57.9	63.0	64.7	51.8	59.3	61.0	64.9	56.0	60.4
FLIPDIR	70.2	69.1	50.0	61.6 (+3.7)	74.4	72.5	45.2	60.8 (+1.5)	70.6	70.9	51.7	63.0 (+2.6)
<i>Robo2VLM</i>	Exposure				Lens smudge				Motion blur			
	C	N	F	H	C	N	F	H	C	N	F	H
Base	100	100	0.0	0.0	100	100	0.0	0.0	100	100	0.0	0.0
VTI	48.3	46.6	32.9	41.4	48.3	48.9	32.7	41.8	48.3	49.1	33.2	42.1
VCD	80.1	79.5	41.6	61.1	80.6	79.7	47.1	64.9	78.8	80.7	44.9	63.4
LEAD	84.5	85.3	56.4	72.7	83.1	83.6	52.9	69.9	84.5	83.0	44.4	64.6
FLIPDIR	88.2	87.8	55.0	73.4 (+0.7)	87.8	86.4	56.2	73.6 (+3.7)	85.7	82.8	50.8	69.1 (+4.5)
<i>GMAI-MMBench</i>	Window level				Defocus blur				Rotation			
	C	N	F	H	C	N	F	H	C	N	F	H
Base	100	100	0.0	0.0	100	100	0.0	0.0	100	100	0.0	0.0
VTI	39.5	41.7	34.3	38.3	39.5	42.9	33.8	38.4	39.5	41.6	35.7	38.8
VCD	82.0	81.9	45.1	64.4	78.5	81.2	42.9	62.0	79.0	83.2	42.1	61.9
LEAD	80.5	76.9	51.0	66.6	80.5	75.9	48.1	64.6	80.5	75.1	44.4	62.2
FLIPDIR	88.5	89.4	49.0	70.0 (+3.4)	82.5	84.3	54.5	70.9 (+6.3)	84.5	83.2	43.7	64.2 (+2.0)

5.1 EXPERIMENTAL SETUP

Models. Since FLIPDIR requires access to intermediate hidden states, we evaluate it on two open-weight VLMs, Gemma-3-12B-IT (Gemma Team et al., 2025) and Qwen3-VL-8B-Instruct (Bai et al., 2025).

Baselines. We compare against the unmodified model (Base), two decoding-time methods, VCD (Leng et al., 2024) and LEAD (Xu et al., 2026), and the activation-steering method VTI (Liu et al., 2025). We follow the recommended configurations for LEAD and VTI; implementation details appear in Appendix C.4.

Metrics. We measure preservation and recovery of the model’s original answers: preservation on clean (C) and non-flip (N) inputs, and recovery on flipped inputs (F). Our primary metric is their harmonic mean,

$$H(C, N, F) = \left[\frac{1}{3} \left(\frac{1}{C} + \frac{1}{N} + \frac{1}{F} \right) \right]^{-1}, \quad (5)$$

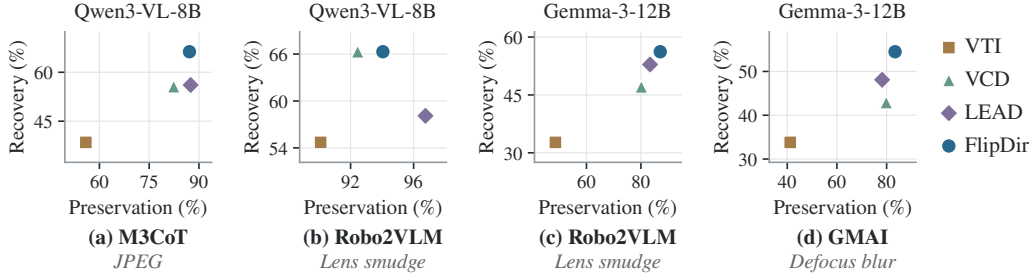


Figure 6: **Recovery–preservation trade-offs in selected settings.** Each point is a reported configuration, not a tuning sweep. All 18 settings, including Base, appear in Appendix E.1.

with $H(C, N, F) = 0$ if any component is zero. Using each model’s original prediction as the reference separates perturbation-induced instability from underlying task accuracy; recovery measures restoration of that prediction rather than ground-truth correctness.

Implementation details. We use rank $k = 8$ and shared gate thresholds $(\tau_o, \tau_c) = (0.3, 0.7)$ across all settings. For each model and perturbation setting, we estimate the flip subspace from the training split and select the steering layer ℓ and strength α on the validation split. The test split is used only for final evaluation. We use deterministic decoding where available for reproducibility. Implementation and selection details appear in Appendix C.

5.2 MAIN RESULTS

FLIPDIR achieves the highest $H(C, N, F)$ in all 18 settings, exceeding the strongest baseline by 2.8 points on average (Tables 3 and 4). Mean gains are 2.5, 2.5, and 3.4 points for science, robotics, and medical VQA, and 2.4 and 3.2 points for Qwen3-VL-8B and Gemma-3-12B, respectively. These gains reflect both preservation and recovery. On Gemma/Robo2VLM exposure, FLIPDIR slightly reduces recovery relative to LEAD (55.0 versus 56.4) but improves C/N from 84.5/85.3 to 88.2/87.8, raising H from 72.7 to 73.4. On Qwen/M3CoT JPEG, it instead raises recovery from the strongest baseline’s 56.1 to 66.3 while keeping C/N above 87%, improving H by 5.1 points.

Figure 6 compares flip recovery (F) with average clean and non-flip preservation, $P = (C+N)/2$, in four settings. Across all 18 settings, FLIPDIR is Pareto-nondominated among the reported operating points, with some baselines also on the frontier (Appendix E.1).

5.3 TRANSFER ACROSS DATASETS AND PERTURBATION COMBINATIONS

We transfer only U across datasets and perturbation combinations, retaining target-specific layers, strengths, and gate thresholds (Figure 7). Across the twelve off-diagonal pairs, H decreases by 0.1 to 6.8 points relative to the target-matched subspace. The largest relative decrease occurs for Robo2VLM lens smudge $\rightarrow$ M3CoT exposure on Qwen (71.5 versus 78.3, retaining 91% of target-matched H). This supports partial transfer under target-specific calibration; dataset and perturbation effects are not isolated. Appendix E.2 gives full results.

Source	Target					
	M3CoT			Robo2VLM		
	Exposure			Lens smudge		
	Exposure	Lens smudge	Rotation	Exposure	Lens smudge	Rotation
M3CoT Exposure	* 61.6	69.4	62.3	* 78.3	75.7	65.3
Robo2VLM Lens smudge	61.5	* 73.6	60.5	71.5	* 82.5	69.7
GMAI Rotation	61.4	71.6	* 64.2	74.2	76.3	* 70.7

(a) Gemma-3-12B

Source	Target					
	M3CoT			Robo2VLM		
	Exposure			Lens smudge		
	Exposure	Lens smudge	Rotation	Exposure	Lens smudge	Rotation
M3CoT Exposure	* 61.6	69.4	62.3	* 78.3	75.7	65.3
Robo2VLM Lens smudge	61.5	* 73.6	60.5	71.5	* 82.5	69.7
GMAI Rotation	61.4	71.6	* 64.2	74.2	76.3	* 70.7

(b) Qwen3-VL-8B

Figure 7: **Flip-subspace transfer (H , %).** Column-wise min-max colors compare sources for each target. White asterisks mark matched source–target pairs.

Source	Target	Rank k				
		2	4	8	16	32
		Robo2VLM Lens smudge	GMAI Rotation	M3CoT Exposure	Robo2VLM Lens smudge	GMAI Rotation
		82.2	79.6	82.5	79.1	77.8
Qwen3-VL-8B	Robo2VLM Lens smudge	82.2	79.6	82.5	79.1	77.8
	GMAI Rotation	65.4	64.7	70.7	63.7	69.6
	M3CoT Exposure	59.2	58.6	61.6	59.2	60.2
	Robo2VLM Lens smudge	77.1	70.4	73.6	71.4	70.2
Gemma-3-12B	GMAI Rotation	65.1	64.2	64.2	65.4	64.5

Figure 8: **Rank sensitivity (H , %).** Row-normalized colors; boxed $k = 8$. GMAI denotes GMAI-MMBench.

5.4 ABLATION STUDIES

Representation difference. Flip-inducing and non-flipping perturbations yield similar mean activation-shift norms (15.2 versus 15.6) but differ in direction, suggesting that *answer flips* are associated with the direction rather than the magnitude of the representation change. On Robo2VLM exposure and GMAI window level, we re-estimate U for Gemma-3-12B from three representation differences while holding all other components fixed (Table 5). *Flip – Clean* achieves the highest H in both evaluations. These results suggest that conditioning on *answer flips* enriches perturbation-induced representation shifts for directions useful for recovery.

Table 5: Representation ablation (H , %).

Representation	Robo2VLM	GMAI
Flip – Clean	73.4	70.0
Non-flip – Clean	61.5	67.9
Flip – Non-flip	68.8	64.6

Table 6: Margin-gate ablation (H , %); Δ = On – Off.

Setting	Gemma-3-12B			Qwen3-VL-8B		
	Off	On	Δ	Off	On	Δ
M3CoT exp.	54.8	61.6	+6.8	75.6	78.3	+2.7
Robo2VLM lens	68.8	73.6	+4.8	73.5	82.5	+9.0
GMAI rot.	58.3	64.2	+5.9	69.2	70.7	+1.5

Margin gate. Table 6 compares margin-gated steering with steering at every decoding step, holding other components fixed within each setting. The gate improves H in all six comparisons. Figure 13 in Appendix F.2 shows gains in C, N, F, and H; Figure 9(a) varies thresholds on Qwen3-VL-8B/M3CoT exposure, with the H axis starting at 74%.

Steering strength. Figure 9(b) varies α on Gemma-3-12B/Robo2VLM exposure. Partial attenuation at $\alpha = 0.5$ achieves higher H than either weaker steering at $\alpha = 0.1$ or full projection removal at $\alpha = 1$. Each sweep evaluates one setting.

Subspace rank. Across the five settings in Figure 8, increasing k beyond 8 yields no consistent gain, although the best rank varies by setting. Additional ablations appear in Appendix F.

Computational efficiency. FLIPDIR estimates the subspace once and reuses its dk coefficients across queries. At matched input/output lengths, conditional recomputation adds approximately 1 to 19% to Base decoding FLOPs in settings with gate traces (Table 7); our current harness instead computes both paths at every step. Appendix G.1 details construction and FLOPs assumptions, and Appendix G.3 reports measured runtime.

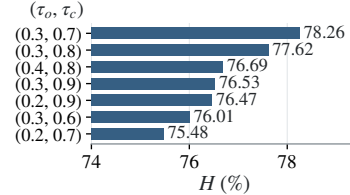
Table 7: Offline requirements and estimated decoding FLOPs relative to Base.

	Base	LEAD	VTI	VCD	FLIPDIR
Offline directions	No	No	Yes	No	Yes
Decode FLOPs	1.00×	≈ 1.08 – $1.09\times$	$\approx 1.00\times$	$\approx 2.00\times$	≈ 1.01 – $1.19\times$

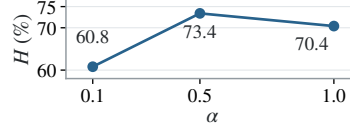
6 CONCLUSION

We study *answer flips* under subtle visual changes common in real-world image capture, processing, and transmission. Observations of activation spectra and token margins motivate FLIPDIR, which selectively attenuates a learned flip subspace during decoding without weight updates or a clean reference. We introduce VISFLIP across scientific reasoning, robot-scene understanding, and medical VQA. Across 18 settings, FLIPDIR improves the harmonic mean of preservation and recovery by 2.8 points on average over the strongest baseline.

Limitations and future work. Our objective is consistency with the model’s original prediction; FLIPDIR does not aim to correct errors already present on clean inputs. Evaluation covers two



(a) Margin gate



(b) Steering strength

Figure 9: **Gate and strength sensitivity.** Other components are fixed.

open-weight VLMs and fixed perturbation families across three domains. Deployment requires hidden-state access and validation-based selection of the steering layer and strength; transfer currently retains target-specific calibration. Future work includes unseen and combined perturbations and reduced calibration requirements.

AI USE STATEMENT

We used generative AI tools to assist with manuscript drafting and proofreading, figure creation and revision, and code development and revision. We also used these tools for feedback on methodological explanations and assistance with interpreting results and estimating computational costs. All AI-assisted work was conducted under detailed guidance from the authors. The authors subsequently reviewed and verified all AI-assisted text, figures, and code for technical accuracy and consistency with the study. We take full responsibility for the final manuscript, including its claims, results, and associated artifacts.

REPRODUCIBILITY STATEMENT

We will publicly release the complete VISFLIP dataset, the code used to construct it, and all experimental code, including implementations of FLIPDIR and the evaluated baselines, evaluation scripts, and analysis and figure-generation scripts. Section 3 describes the method, and Section 4 and Appendix A document benchmark construction, perturbation settings, and data splits. Appendix B specifies the evaluation pools and budget-matching protocol. Appendix C provides implementation details, hyperparameter selection, baseline adaptations, and inference settings. The release will also include versioned configurations, random seeds and provenance metadata, and the extracted flip subspaces, as described in Appendix C.5. Computational cost accounting is provided in Appendix G.1.

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A BENCHMARK CONSTRUCTION AND DATA

A.1 BENCHMARK CONSTRUCTION

Algorithm 1 constructs model-specific answer-flip instances. We first obtain a greedy clean-image answer and exclude parents whose answer cannot be normalized to the option space. Perturbations are then sampled from the fixed ranges in Table 9 using a per-parent seed chain and evaluated with the same prompt. Each child is labeled relative to its parent’s clean prediction using Eq. 1. Full-budget runs evaluate all B draws; pair-existence runs may stop once both a flip and a non-flip child are found. The released field `stable` denotes non-flip children. Seeds, parameters, prompts, outputs, and parent identifiers are retained for reconstruction.

Algorithm 1 Answer-flip benchmark construction (full-budget variant)

Require: Base dataset $\mathcal{D}$, VLM f , perturbation family $\mathcal{P}$, budget B

```

1: for all  $(I, T, y^*) \in \mathcal{D}$  do
2:    $y_{\text{clean}} \leftarrow f(I, T)$ 
3:   if  $y_{\text{clean}}$  cannot be normalized then
4:     continue
5:   end if
6:   for  $b = 1$  to  $B$  do
7:     Sample  $\tau_b \sim \mathcal{P}$  using a deterministic seed chain
8:      $\tilde{I}_b \leftarrow \tau_b(I)$ ;  $\tilde{y}_b \leftarrow f(\tilde{I}_b, T)$ 
9:      $z_b \leftarrow \mathbb{I}[\text{norm}(\tilde{y}_b) \neq \text{norm}(y_{\text{clean}})]$ 
10:    Store image, seed, variation parameters, raw output, normalized answer, and label  $z_b$ 
11:   end for
12: end for
```

Pair-search cost. Pair search stops when both child types are found, after 15 draws without a flip, or at the 50-draw cap. This stopping rule differs from the full-budget procedure in Algorithm 1.

Table 8 reports the number of VLM calls under this stopping rule, including the clean call and subsequent perturbed-image calls. Each search is defined by a parent, a perturbation family, and a backbone; the same source parent may therefore contribute to multiple searches. Across 27,478 searches, construction uses 352,970 calls: 87,384 for searches that obtain both child types and 265,586 for those that do not. Counts include all calls made within each search, not only the retained children.

Table 8: **VLM calls for benchmark construction.** Calls per search are reported as mean $\pm$ standard deviation. Pair found denotes obtaining both a flip and a non-flip child.

Domain	Pair found			Pair not found			All calls
	Searches	Calls/search	Calls	Searches	Calls/search	Calls	
<i>Gemma-3-12B</i>							
M3CoT	3,817	6.25 ± 5.08	23,858	2,276	16.82 ± 5.28	38,271	62,129
Robo2VLM	1,485	6.99 ± 5.09	10,387	4,716	16.10 ± 1.90	75,946	86,333
GMAI-MMBench	781	6.43 ± 5.00	5,021	719	17.02 ± 5.90	12,239	17,260
<i>Qwen3-VL-8B</i>							
M3CoT	2,971	9.71 ± 7.14	28,847	3,062	16.45 ± 3.93	50,357	79,204
Robo2VLM	1,312	10.02 ± 7.46	13,148	4,863	16.11 ± 1.94	78,333	91,481
GMAI-MMBench	876	6.99 ± 5.22	6,123	600	17.40 ± 6.86	10,440	16,563
All	11,242	7.77 ± 6.21	87,384	16,236	16.36 ± 3.52	265,586	352,970

A.2 PERTURBATION DEFINITIONS AND RANGES

Perturbation operators. Exposure adjusts image intensity in EV units; white balance applies a global illuminant shift (Afifi et al., 2019); double JPEG recompresses the image twice at distinct quality levels. Lens smudge composites one to two Bezier wipe smears (opacity 0.28 to 0.45), two to four droplets (0.30 to 0.50), and two to six dust specks, with local blur ($\sigma = 4.5$), 22% haze, and an opacity cap of 0.6 (Park et al., 2026). Motion blur uses an inertial camera-shake trajectory at

Table 9: **Perturbation ranges.** Each instance uses a deterministic seed chain. Appendix A.5 specifies modality-dependent medical operators.

Domain	Variation	Sampling band
M3CoT	Exposure shift	$ \text{EV} \in [0.45, 0.65]$
M3CoT	White-balance shift	1000 to 3000 K
M3CoT	Double JPEG	$q_1 \in [85, 95], q_2 \in [72, 85], q_2 \neq q_1$
Robo2VLM	Exposure shift	$ \text{EV} \in [0.45, 0.65]$
Robo2VLM	Lens smudge	wipe smears, droplets, dust; opacity ≤ 0.6 , blur $\sigma = 4.5$
Robo2VLM	Motion blur	inertial trajectory, intensity 0.35
GMAI-MMBench	Window-level shift	contrast and gamma in $[0.80, 1.25]$
GMAI-MMBench	Defocus blur	modality-gated PSF (disk or Gaussian)
GMAI-MMBench	Detector rotation	$\pm 1.5^\circ$ to 4° , reflect-pad and center crop

intensity 0.35. Medical operators are specified in Appendix A.5. Reported ΔE is the mean per-pixel CIE76 difference in CIELAB space.

A.3 DATASET COMPOSITION AND TASK COVERAGE

Table 2 summarizes the parent-image pools used in the main benchmark. M3CoT contributes eight physics topics covering scientific diagrams and multi-step conceptual reasoning. Robo2VLM contributes eight question types spanning robot state, reachability, spatial correspondence, depth, action outcomes, and grasping. GMAI-MMBench is organized by clinical VQA task, perceptual granularity, and imaging modality, with both lens-based and detector-based images represented. Parents in the training split serve as offline calibration data to estimate U , validation parents to select the layer ℓ and strength α , and test parents for final evaluation. These pools are disjoint. Validation selection uses parents with both a flip and a non-flip child; its effective size therefore depends on the setting (Appendix C.3).

A.4 SPLIT CONSTRUCTION AND FILTERING

For M3CoT and Robo2VLM, we retain the domain and task definitions of the source datasets and apply model-specific usability filtering only after obtaining the clean prediction. Parents whose clean answer cannot be normalized to the task answer space are excluded, which is why the usable evaluation counts can differ slightly across models (Table 10).

For GMAI-MMBench, we filter the public validation release and partition it with fixed-seed largest-remainder allocation over joint clinical-task and perceptual-granularity strata. Task, granularity, and modality marginals differ by about two percentage points or less between the resulting train and test pools. Appendix A.5 provides the modality-specific filtering and image-processing details.

A.5 MEDICAL DATA AND PREPROCESSING

We remove GMAI-MMBench questions with fewer than four options or answerable by a single text-only greedy query, leaving 2,397 candidates. Lens-based modalities (endoscopy, fundus, dermoscopy, and microscopy) use a disk point-spread function for defocus (Goodman, 2005; Mannan & Langer, 2016); X-ray and CT use a Gaussian kernel (Bushberg et al., 2020). MRI is outside this isotropic-blur protocol. Window-level shift composes contrast and gamma changes in $[0.80, 1.25]$. Rotation uses reflect padding and center cropping to model small in-plane alignment changes.

Images are downsampled, when needed, to a maximum long side of 1280 pixels for Qwen3-VL and 896 for Gemma-3, following the VLMEvalKit convention. Perturbations are applied after resizing, with blur radius defined relative to the short side. Table 10 reports the model-specific subspace and evaluation pools.

B EVALUATION PROTOCOL AND ANSWER-FLIP STATISTICS

B.1 EVALUATION POOL SIZES

Table 10 lists the calibration pairs and the actual evaluation denominators for each setting.

Table 10: **Subspace and evaluation pool sizes.** U pairs counts the pairs used to estimate the subspace, at most one per parent. Eval C/N/F lists the actual clean, non-flip, and flip denominators used for recovery metrics. Parent pools are given in Table 2.

Setting	Qwen3-VL-8B		Gemma-3-12B	
	U pairs	Eval C/N/F	U pairs	Eval C/N/F
M3CoT exposure	737	429/428/183	989	449/447/278
M3CoT white balance	723	429/428/190	977	449/448/272
M3CoT JPEG	759	429/428/187	1039	449/444/298
Robo2VLM exposure	334	560/558/107	316	567/566/149
Robo2VLM lens smudge	255	560/554/86	319	567/566/153
Robo2VLM motion blur	287	560/558/94	375	567/564/187
GMAI window level	183	192/190/112	185	200/199/102
GMAI defocus blur	146	192/189/85	120	200/191/77
GMAI detector rotation	185	192/192/110	192	200/197/126

B.2 BUDGET-MATCHED CROSS-MODEL EVALUATION

Cross-model comparisons use a common budget of $B = 15$ perturbation draws per parent. For open-weight models, recorded children are matched to reconstructed draws with a tolerance of two pixel-intensity levels, and only the first 15 draws are counted. API models use this budget directly. These rates use a separate parent pool from the test-split percentages in Table 11.

Table 1 reports clean accuracy and the *answer-flip rate* under this protocol. All four models exhibit flips, including the two closed models. In this comparison, flip rate decreases as clean accuracy increases, but not with parameter count. On flipped inputs, the two open-weight models achieve only about 30% accuracy, showing that most flips are errors rather than harmless rephrasings. Flips also occur on many of the same inputs across models. A question that flips Qwen3-VL-8B is more likely to flip Claude Opus 4.8 (odds ratio 5.2, 95% CI [3.1, 8.8]), and 31 of GPT-5.5’s 41 flips occur on parents that also flip Qwen3-VL-8B. These results indicate overlap in the inputs on which different models exhibit answer instability.

B.3 ANSWER-FLIP RATES ACROSS ALL EVALUATION SETTINGS

Table 11 reports answer-flip percentages in the evaluation pools, using the clean and flip counts in Table 10. Cross-model comparisons use the separate matched-budget protocol above.

Table 11: **Answer flips in the evaluation pools.** Percentages are $100n_F/n_C$, using the flip and clean counts in Table 10; $n = n_C$.

Setting	Qwen3-VL-8B	Gemma-3-12B
M3CoT exposure	42.7% (n=429)	61.9% (n=449)
M3CoT white balance	44.3% (n=429)	60.6% (n=449)
M3CoT JPEG	43.6% (n=429)	66.4% (n=449)
Robo2VLM exposure	19.1% (n=560)	26.3% (n=567)
Robo2VLM lens smudge	15.4% (n=560)	27.0% (n=567)
Robo2VLM motion blur	16.8% (n=560)	33.0% (n=567)
GMAI window level	58.3% (n=192)	51.0% (n=200)
GMAI defocus blur	44.3% (n=192)	38.5% (n=200)
GMAI detector rotation	57.3% (n=192)	63.0% (n=200)

C IMPLEMENTATION AND REPRODUCIBILITY

C.1 FLIPDIR CONFIGURATION

We select the steering layer and strength separately for each model and perturbation setting. Table 12 lists the selected values and corresponding main-table results. All settings share rank $k = 8$ and gate thresholds $(\tau_o, \tau_c) = (0.3, 0.7)$.

Table 12: **Selected configurations.** Layer ℓ , strength α , and main-table $H(C, N, F)$. All settings use $k = 8$, gate thresholds $(0.3, 0.7)$, and uncentered SVD with at most one flip pair per parent.

Setting	Gemma-3-12B			Qwen3-VL-8B		
	ℓ	α	$H(C, N, F)$	ℓ	α	$H(C, N, F)$
M3CoT exposure	12	1.0	61.6	4	1.0	78.3
M3CoT white balance	12	0.5	60.8	36	1.0	76.4
M3CoT JPEG	12	1.0	63.0	36	1.0	78.9
Robo2VLM exposure	12	0.5	73.4	36	1.0	82.4
Robo2VLM lens smudge	36	0.5	73.6	36	1.0	82.5
Robo2VLM motion blur	36	1.0	69.1	36	0.5	82.4
GMAI window level	12	0.5	70.0	12	0.5	70.1
GMAI defocus blur	36	1.0	70.9	36	1.0	63.7
GMAI rotation	36	0.5	64.2	36	0.5	70.7

C.2 STEERING IMPLEMENTATION

We extract the states used to estimate U at the final prompt position in unedited prefill, but apply the resulting directions only during decoding at the selected layer. The projection can be evaluated as $U(U^\top h)$ using two matrix-vector products, requiring $O(dk)$ arithmetic without materializing a $d \times d$ projection matrix. The evaluation harness computes both the unedited and steered passes at each decoding step, then uses the gate to select the output distribution.

C.3 HYPERPARAMETER SELECTION

The six (ℓ, α) candidates are scored on validation triplets containing a clean image, one flip child, and one non-flip child. Parents follow a fixed 40-shard ordering; the first R shards are used to cover approximately 50 flip-bearing parents, with R adjusted to the setting’s flip density. Qwen3-VL defocus blur has 47 available validation flip pairs and approximately 24 selection triplets. Validation parents are disjoint from subspace-calibration and test parents. Varied images are stored as PNG at construction time and are regenerable exactly from the deterministic per-parent seed chain.

C.4 BASELINE ADAPTATIONS

We retain the original LEAD constants and official VTI strengths rather than retuning them for individual benchmark settings. The adaptations below address backbone compatibility. In contrast, FLIPDIR selects its method-specific intervention layer and strength on held-out validation data (Appendix C.3); test data are used only for final evaluation.

LEAD. We adapt the public LEAD implementation for Qwen2.5-VL (Xu et al., 2026) to each model’s embedding and token conventions. For Gemma-3, direct embedding inputs retain the model’s $\sqrt{d}$ scaling, termination recognizes its chat end token, and image-anchor lookup uses the model-specific image token. Table 4 reports this adapted implementation.

VTI. We use the official VTI implementation. Qwen3-VL produces a variable visual-token grid, so we average VTI’s visual direction over tokens within each layer and apply it at all positions. The textual direction is unchanged. Setting $\alpha = \beta = 0$ reproduces the base model exactly.

C.5 DETERMINISTIC INFERENCE AND RELEASED ARTIFACTS

All open-weight experiments use greedy decoding, batch size one, fixed seeds, deterministic kernels, and TF32 disabled. Repeating one complete setting reproduced all 1,040 generations exactly. Exact

reproduction uses the saved flip subspaces: GPU re-extraction can introduce numerical differences, and subspaces agreeing to 10^{-6} can still change H by several points at near-tied decoding steps.

Release contents. We will release model-specific benchmark instances, perturbation code and versioned configurations, inference and evaluation scripts, answer-normalization and flip-mining utilities, budget-matching tools, and extracted subspaces. Instance metadata includes the parent identifier, seed, perturbation parameters, prompt, raw output, and normalized answer.

D ACTIVATION AND DECODING ANALYSES

D.1 ACTIVATION ENERGY CONCENTRATION

For Qwen3-VL-8B on Robo2VLM lens smudge, we analyze 255 clean-flip pairs ($d = 4096$) at layers 4, 12, and 36. The Gaussian null uses a matrix of the same size and the same uncentered SVD and energy calculation. With uncentered singular values $\sigma_{\ell,j}$, cumulative energy is $E_\ell(k) = \sum_{j=1}^k \sigma_{\ell,j}^2 / \sum_j \sigma_{\ell,j}^2$. The top 8 directions capture 48.6 to 53.7% of the energy, versus 4.7% for the Gaussian null; the top 16 capture 59.4 to 65.7%, versus 9.2%. Figures 4 and 10a plot these cumulative energies.

The cumulative-energy plots use a square-root rank axis and a linear energy axis.

D.2 TOKEN MARGINS AT DECODING DIVERGENCE

Figure 10(b) complements the Qwen3-VL-8B results in Figure 5 with Gemma-3-12B. We analyze stored unedited traces for the settings with available traces. At each step, original and perturbed inputs share the same decoding prefix. A divergence occurs when their next-token predictions differ; after each divergence, we advance both runs with the same token to realign the prefix before continuing. Margins are computed from the perturbed-input distribution. Within each backbone, divergence and non-divergence token distributions are normalized separately to unit area. We use Gaussian kernel density estimation with bandwidth 0.045 and reflection at 0 and 1. The square-root density axis makes overlap visible; tick labels report density in the original units.

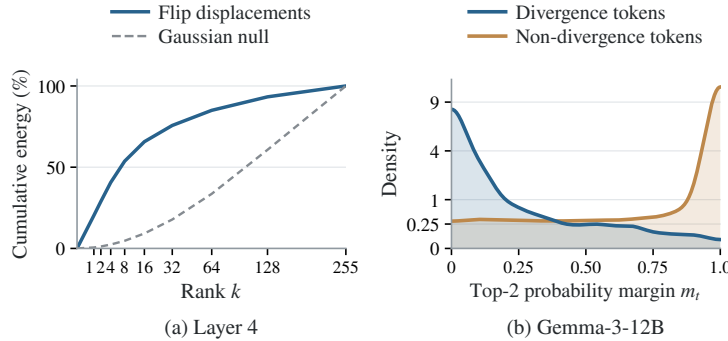


Figure 10: **Additional activation and decoding analyses.** (a) Cumulative energy at layer 4 for the same 255 Qwen3-VL-8B/Robo2VLM lens-smudge pairs as Figure 4. (b) Divergence and non-divergence token margins for Gemma-3-12B, complementing the Qwen results in Figure 5.

D.3 LIMITS OF STATIC DIAGNOSTICS

We examine whether simple diagnostics predict *answer flips* or recovery. Subspace alignment has little correlation with realized recovery ($\rho \approx 0$), and steering coordinates provide weak token-level discrimination (SNR 0.02 to 0.07; AUC 0.55). A parent’s flip probability is also nearly flat in its clean-run margin (0.48 to 0.60). These results limit the use of alignment, steering coordinates, or clean-run confidence as standalone predictors. FLIPDIR does not use U as a flip detector or clean-run confidence to predict future flips. It uses U as an intervention subspace and gates steering with the current unedited token margin during decoding of the perturbed input (Section 3). The diagnostics therefore distinguish static prediction from decoding-time intervention; they do not by themselves establish the benefit of dynamic gating. That benefit is evaluated directly by the controlled gate

ablation (Table 6). We select (ℓ, α) using held-out validation performance rather than these proxies (Appendix C.3).

E ADDITIONAL EVALUATION RESULTS

E.1 RECOVERY-PRESERVATION TRADE-OFFS

We plot $P = (C + N)/2$ against flip recovery F for all 18 settings in Tables 3 and 4. Each point represents the reported configuration, without additional tuning or interpolation. A point is Pareto-nondominated if no compared configuration has at least as high P and F and is strictly higher in one. FLIPDIR is nondominated in every setting under the reported rounded values, although other methods can remain on the frontier. Thus these comparisons show trade-offs independently of H , not dominance at every preservation level.

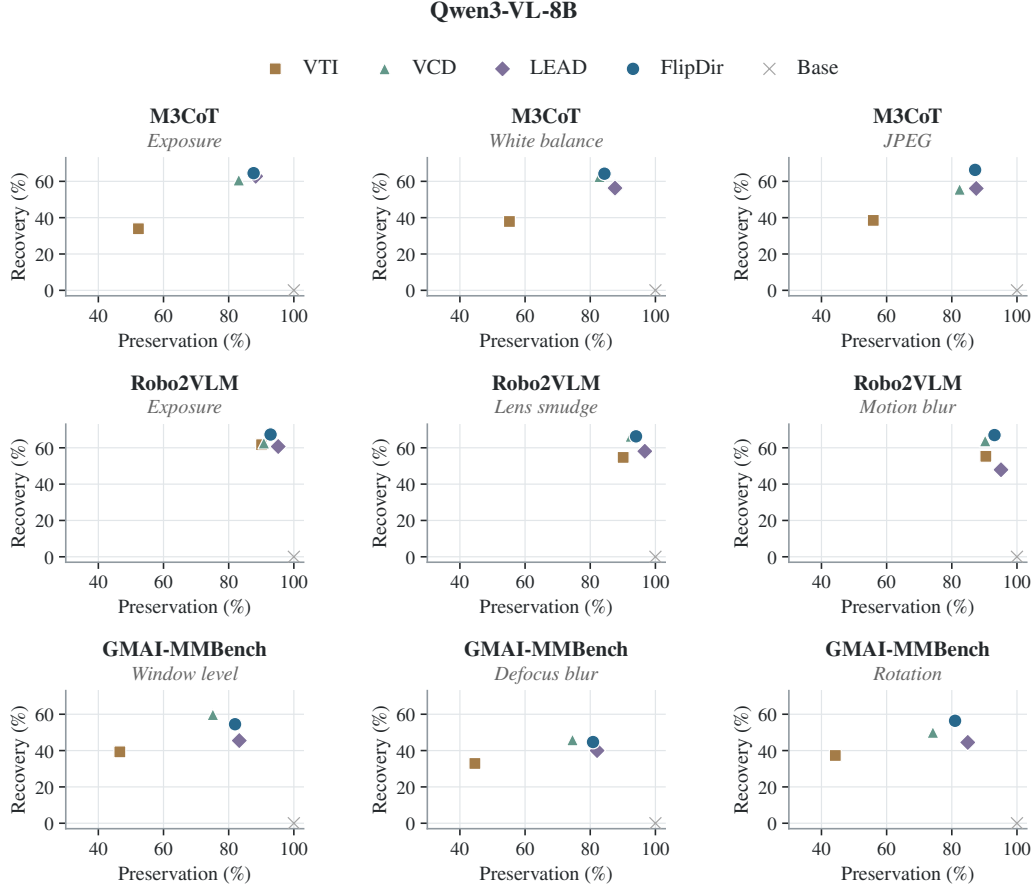


Figure 11: **Recovery–preservation trade-offs for Qwen3-VL-8B.** All nine settings and five methods are shown on shared axes. Higher and further right indicate better recovery and preservation, respectively.

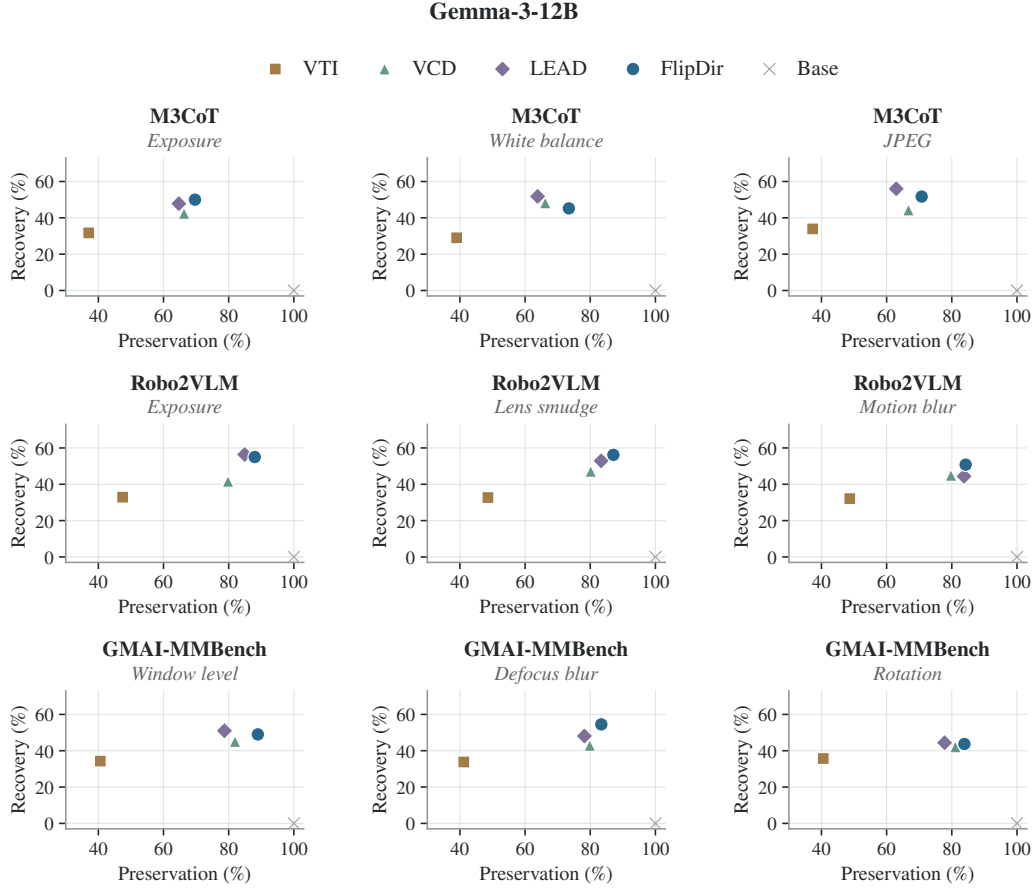


Figure 12: **Recovery–preservation trade-offs for Gemma-3-12B.** All nine settings, with the same methods and axes as Figure 11. Base is at (100, 0) in every setting.

E.2 TRANSFER ACROSS DATASETS AND PERTURBATIONS

Table 13 expands Figure 7 with all measured source-to-target transfers, including within-domain sources. Only U is transferred; the layer, strength, and gate thresholds use the target-specific configuration. Across the cross-domain combinations shown in Figure 7, at least 91% of target-matched H is retained.

Table 13: **Transfer results.** $H(C, N, F)$ using the source subspace and the target-specific layer, strength, and gate thresholds. The last column uses a target-matched subspace.

Model	Target setting	Source U	$H(C, N, F)$	Target-specific
Qwen3-VL-8B	Robo2VLM lens	M3CoT exposure	75.7	82.5
Qwen3-VL-8B	Robo2VLM lens	GMAI detector	76.3	82.5
Qwen3-VL-8B	M3CoT exposure	Robo2VLM lens	71.5	78.3
Qwen3-VL-8B	M3CoT exposure	GMAI detector	74.2	78.3
Qwen3-VL-8B	GMAI detector	M3CoT exposure	65.3	70.7
Qwen3-VL-8B	GMAI detector	Robo2VLM lens	69.7	70.7
Gemma-3-12B	M3CoT exposure	Robo2VLM lens	61.5	61.6
Gemma-3-12B	M3CoT exposure	GMAI detector	61.4	61.6
Gemma-3-12B	Robo2VLM lens	M3CoT exposure	69.4	73.6
Gemma-3-12B	Robo2VLM lens	GMAI detector	71.6	73.6
Gemma-3-12B	GMAI detector	M3CoT exposure	62.3	64.2
Gemma-3-12B	GMAI detector	Robo2VLM lens	60.5	64.2
Gemma-3-12B	M3CoT exposure	M3CoT JPEG	61.6	61.6
Gemma-3-12B	M3CoT exposure	Robo2VLM exposure	61.1	61.6
Gemma-3-12B	Robo2VLM lens	Robo2VLM exposure	71.1	73.6
Gemma-3-12B	GMAI detector	GMAI window	61.1	64.2
Qwen3-VL-8B	GMAI detector	GMAI window	60.9	70.7
Gemma-3-12B	Robo2VLM lens	Robo2VLM motion	71.6	73.6
Gemma-3-12B	GMAI detector	Robo2VLM motion	62.1	64.2
Qwen3-VL-8B	Robo2VLM lens	Robo2VLM motion	80.9	82.5
Qwen3-VL-8B	GMAI detector	Robo2VLM motion	70.2	70.7

F ABLATIONS AND SENSITIVITY

This section supplements the rank, gate, and strength analyses in Section 5.4 with tabulated rank results, the full margin-gate comparison, and sensitivity to the number of calibration pairs.

F.1 FLIP-SUBSPACE RANK

Table 14 sweeps k on six settings, covering two non-medical perturbation families and one medical perturbation family per model, while holding all other components fixed at the selected configuration.

Table 14: **Rank sensitivity.** $H(C, N, F)$ with all other components fixed to Table 12. The shared rank $k = 8$ is shaded; row maxima are bold.

Model	Setting	$k = 2$	4	8	16	32
Qwen3-VL-8B	M3CoT exposure	71.0	74.3	78.3	73.2	72.6
	Robo2VLM lens smudge	82.2	79.6	82.5	79.1	77.8
	GMAI rotation	65.4	64.7	70.7	63.7	69.6
Gemma-3-12B	M3CoT exposure	59.2	58.6	61.6	59.2	60.2
	Robo2VLM lens smudge	77.1	70.4	73.6	71.4	70.2
	GMAI rotation	65.1	64.2	64.2	65.4	64.5

F.2 MARGIN-GATE ABLATION

Table 15 reports the preservation and recovery metrics underlying Table 6. Gate Off applies steering at every decoding step; the subspace, layer, and strength are held fixed within each comparison. The gate improves H in all six settings. Figure 13 visualizes the corresponding changes in each metric.

Table 15: **Margin-gate ablation.** C/N/F/H (%) with and without the gate. $\Delta H = H_{\text{On}} - H_{\text{Off}}$, computed from the displayed values.

Setting	Gate Off				Gate On				
	C	N	F	H	C	N	F	H	ΔH
<i>Gemma-3-12B</i>									
M3CoT exposure	59.5	63.8	45.0	54.8	70.2	69.1	50.0	61.6	+6.8
Robo2VLM lens smudge	81.7	83.4	51.6	68.8	87.8	86.4	56.2	73.6	+4.8
GMAI rotation	84.0	78.7	37.3	58.3	84.5	83.2	43.7	64.2	+5.9
<i>Qwen3-VL-8B</i>									
M3CoT exposure	84.1	85.3	62.3	75.6	87.4	87.9	64.5	78.3	+2.7
Robo2VLM lens smudge	92.3	91.9	52.3	73.5	94.1	94.0	66.3	82.5	+9.0
GMAI rotation	79.2	81.8	53.6	69.2	80.2	81.8	56.4	70.7	+1.5

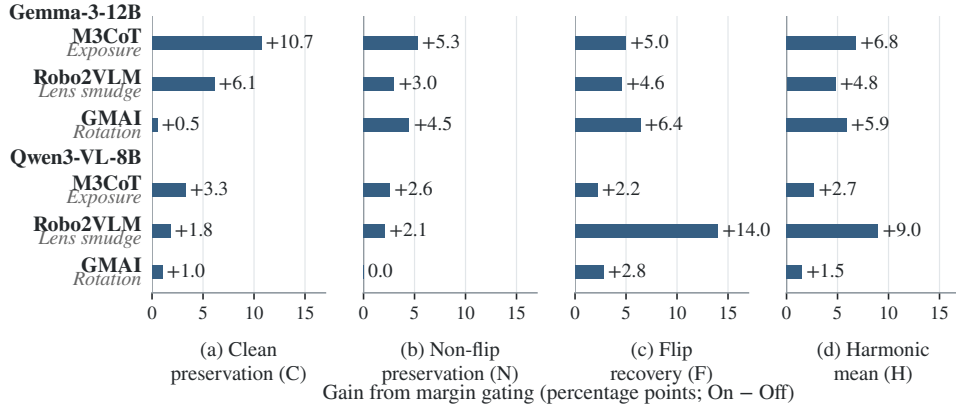


Figure 13: **Performance gains from margin gating.** Changes in clean preservation (C), non-flip preservation (N), flip recovery (F), and their harmonic mean (H), measured in percentage points as Gate On minus Gate Off. All panels share the same scale; values are computed from Table 15. The subspace, layer, and strength are fixed within each comparison. GMAI denotes GMAI-MMBench.

F.3 CALIBRATION SAMPLE SIZE

Table 16 varies the number of pairs used to estimate U , with other components fixed. Each subset contains the first N parents in sorted identifier order, with at most one flip pair per parent; the basis is estimated by uncentered SVD.

Table 16: **Subspace-estimation sample size.** $H(C, N, F)$ with other components fixed; parent counts are shown in parentheses.

Setting	50 pairs	subsampled	full pool
Gemma M3CoT exposure	60.2	56.8 (250)	61.6 (989)
Gemma Robo2VLM lens	67.0	72.2 (150)	73.6 (319)
Gemma GMAI detector	67.0	63.8 (100)	64.2 (192)
Qwen Robo2VLM lens	78.6	79.2 (150)	82.5 (255)

G COMPUTE, RUNTIME, AND GATE STATISTICS

G.1 COMPUTE ACCOUNTING

Offline direction construction. Direction estimation is performed once per backbone and perturbation setting, and the resulting subspace is reused across queries. Given existing clean and flip pairs, activation extraction requires two prefill-only forwards per pair; the candidate layers are read from the same forwards. The 4,512 Gemma pairs and 3,609 Qwen pairs in Table 10 therefore require 9,024 and 7,218 prefills, respectively. SVD is then applied to the activation differences, and the retained basis stores dk coefficients. These extraction counts exclude pair search and validation-based parameter selection. Pair-search costs are reported separately in Table 8; validation selects the steering layer and strength before deployment. Thus the method incurs an initial construction cost in exchange for a reusable intervention, without updating model weights.

Decoding FLOPs. We count a multiply-add as two FLOPs and approximate the dominant linear operations in the decoder blocks and output head. Comparisons fix the input and output lengths and exclude prefill, image encoding, and the context-length-dependent attention products. Let F_b be the per-token cost of one decoder block, F_h the output-head cost, and L the number of blocks. Standard decoding costs $F_0 = LF_b + F_h$. Under conditional execution, an open gate triggers recomputation only above the intervention layer ℓ , including the output head. The relative cost is

$$\frac{F_{\text{FlipDir}}}{F_0} \approx 1 + p_{\text{open}} \frac{(L - \ell)F_b + F_h + 4dk}{LF_b + F_h}. \quad (6)$$

The projection $U(U^\top h)$ costs approximately $4dk$ FLOPs; gate arithmetic is negligible at this scale. The output head is recomputed even when $\ell = L$, so intervention at the final layer does not have zero cost.

For Gemma-3-12B, $L = 48$, $d = 3840$, $F_b \approx 4.48 \times 10^8$, and $F_h \approx 2.01 \times 10^9$ FLOPs. For Qwen3-VL-8B, $L = 36$, $d = 4096$, $F_b \approx 3.86 \times 10^8$, and $F_h \approx 1.24 \times 10^9$. We combine the layers in Table 12 with the gate-open fractions replayed on stored unedited traces in Table 17. Estimated multipliers range from 1.068 to 1.185 for Gemma and 1.011 to 1.176 for Qwen, across seven and eight settings with available traces, respectively. These replay fractions are proxies for conditional execution, rather than measurements from a deployed conditional implementation.

LEAD uses one decoder forward and a dense probability-weighted embedding, $\sum_{v=1}^V p_t(v)e_v$, at each step. This adds approximately $2Vd$ FLOPs, giving $F_{\text{LEAD}}/F_0 \approx 1 + 2Vd/F_0$: $1.086\times$ for Gemma and $1.082\times$ for Qwen. VTI uses one forward plus additive edits, giving approximately $1\times$. VCD uses two forwards, giving approximately $2\times$. Under the same accounting, conditional FLIPDIR uses about 41 to 50% fewer decoding FLOPs than VCD. This comparison concerns decoding arithmetic rather than end-to-end latency. Our current evaluation harness computes both the unedited and steered paths at each step, so Table 7 describes the conditional execution of the method rather than measured FLOPs of that harness. Section G.3 reports its measured wall-clock runtime separately.

G.2 GATE ACTIVATION STATISTICS

Gate open rates replay the selected thresholds (0.3, 0.7) with hysteresis over stored unedited traces: twelve parents per setting across shards, including clean, flip, and non-flip generations. N/A marks settings without stored traces.

Table 17: **Runtime and gate statistics.** Time is measured in seconds per parent under the shared serving regime. Gate statistics are replayed on stored unedited traces; entries are counted per 100 tokens. N/A denotes unavailable measurements.

Setting	Gate		Seconds per parent			
	Open %	Entries	FLIPDIR	VCD	LEAD	VTI
Qwen M3CoT exposure	15.5	9.1	254.1	260.5	92.4	252.2
Qwen M3CoT wb	14.5	8.5	231.3	265.4	135.1	227.2
Qwen M3CoT JPEG	12.9	8.0	232.7	258.7	134.6	241.4
Qwen Robo2VLM exp.	17.3	9.6	121.8	109.0	71.9	44.7
Qwen Robo2VLM lens	N/A	N/A	71.8	96.0	49.2	43.7
Qwen Robo2VLM motion	18.8	10.2	112.7	120.4	70.8	42.1
Qwen GMAI window	25.4	13.0	85.7	209.8	113.8	88.6
Qwen GMAI defocus	24.1	12.2	74.5	209.7	114.2	80.5
Qwen GMAI rotation	24.0	12.7	76.8	211.2	115.7	82.8
Gemma M3CoT exposure	13.3	6.5	175.3	168.7	N/A	133.8
Gemma M3CoT wb	11.0	5.9	171.1	172.5	N/A	121.3
Gemma M3CoT JPEG	N/A	N/A	176.3	169.2	96.6	124.3
Gemma Robo2VLM exp.	22.3	11.6	48.0	44.1	N/A	40.8
Gemma Robo2VLM lens	N/A	N/A	36.4	35.5	17.7	38.3
Gemma Robo2VLM motion	24.4	10.8	22.3	44.0	N/A	53.2
Gemma GMAI window	24.0	10.9	53.6	97.5	N/A	89.1
Gemma GMAI defocus	21.7	11.8	49.2	92.6	N/A	74.5
Gemma GMAI rotation	25.8	12.9	54.4	101.5	N/A	91.0

G.3 RUNTIME

Wall-clock time in Table 17 is averaged over the main-table runs under a shared serving regime with four to five concurrent processes per GPU. The current FLIPDIR harness executes both the original and steered passes at each decoding step. Across the six medical settings, the ratio of VCD runtime to FLIPDIR runtime ranges from 1.8 to 2.8, with higher H for FLIPDIR in each setting. These measurements describe the shared serving regime; they do not isolate per-query latency or the conditional-execution FLOPs in Eq. 6.

H QUALITATIVE EXAMPLES

H.1 ANSWER-FLIP RECOVERY

Figure 14 illustrates Gemma-3-12B on Robo2VLM exposure shift with $\ell = 12$ and $\alpha = 0.5$.



Clean image (model answers B)



Exposure shift (−EV)

[Question] *The robot is tasked to moving away with open gripper after releasing the can. After moving away with open gripper after releasing the can, what will be the robot's NEXT action phase?*

- (A) Moving away with open gripper after releasing the can **(B) Approaching the can with open gripper**
 (C) Firmly grasping the can (D) Releasing the can by opening gripper (E) Closing gripper to grasp the can

Varied image, no steering → (A)

The question states that the robot has already “Moving away with open gripper after releasing the can”. The question asks for the *next* action. Since the can has already been released, **the next logical action would be to move away with the open gripper.**

FinalAnswer: (A)

Varied image, FLIPDIR → (B)

The question states that the robot has already “Moving away with open gripper after releasing the can”. This means the can has been released and the robot is moving away. **The next logical action would be to approach another object or location.**

Therefore, the next action phase is approaching the can with an open gripper.

FinalAnswer: (B)

Figure 14: A repaired flip on Gemma-3-12B Robo2VLM exposure shift. The unedited pass on the varied image restates the completed phase and flips to (A); with FLIPDIR the reasoning advances to the cycle’s next phase and the answer returns to (B), the clean answer and the ground truth. Colored spans mark where the trajectories diverge.

H.2 PERTURBATION INTENSITY EXAMPLES

Figures 15, 16, and 17 provide representative examples from the fixed perturbation bands. Additional medical-specific design details are provided in Appendix A.5.



Figure 15: M3CoT variation intensities on the example of Figure 2, rendered from the fixed bands. Rows: exposure shift at the EV band endpoints (+0.45, +0.65), white-balance shift at +1000 K and +3000 K, and the mildest ($q = 95, 85$) and strongest ($q = 85, 72$) double-JPEG draws.

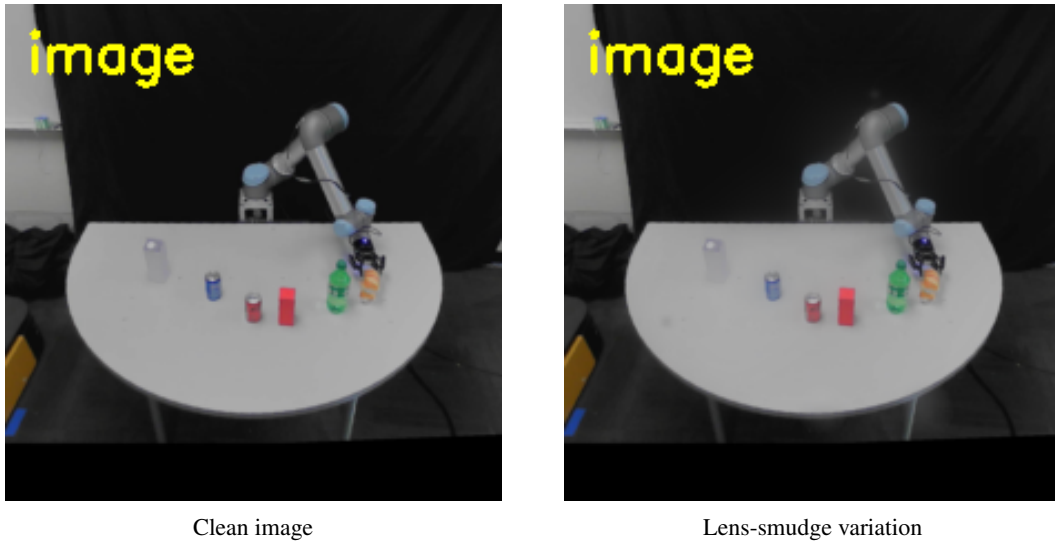


Figure 16: Robo2VLM lens-smudge example from VISFLIP. The perturbation composites wipe smears, droplets, dust, local blur, and mild haze while preserving the underlying scene.

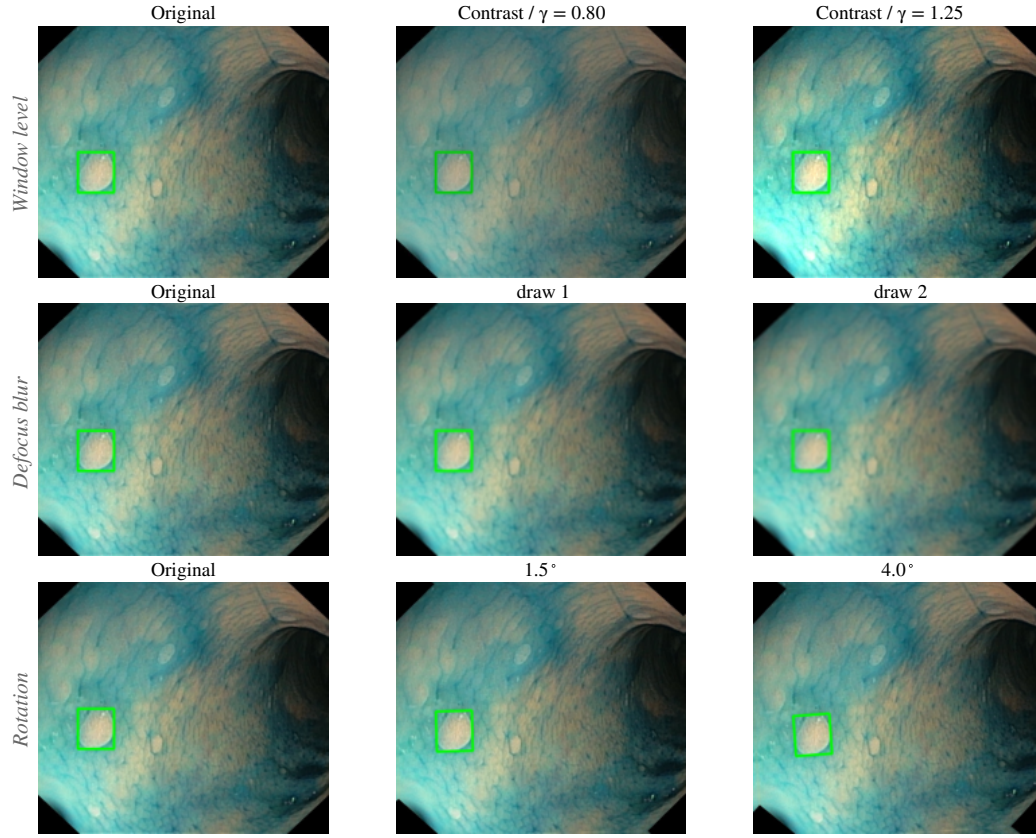


Figure 17: GMAI-MMBench variation intensities on an endoscopy image. Rows: window-level shift at the joint contrast and gamma endpoints, modality-gated defocus blur (disk PSF for this lens modality), and detector rotation at 1.5° and 4.0°.